STUDENT COMMUTE ANALYSIS

Exploratory Data Analysis — End Semester Project

EnTC A3 | Group 1 | B.Tech Semester II

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1. Introduction

Commuting to college is something most of us just deal with — we wake up, figure out how to get there, and somehow make it on time. But for a lot of engineering students in Pune, it is actually a bigger problem than it sounds. We approximately stay about 25 km away from our college, and getting there every single day is genuinely exhausting sometimes. Whether it is waiting for a bus that is never on schedule, or spending almost an hour stuck in traffic, the commute takes up a surprising chunk of the day.

That is kind of where this whole project started. One of the contributors' wondered if other students were dealing with the same issues — long travel times, high costs, or just the general unpredictability of it all. So we decided to actually look into it properly, collect data from fellow students, and do a real analysis on it.

Pune is a growing city and it has a huge student population. Places like Kothrud, Baner, and Wakad are all fairly residential areas, but the colleges and tech parks they commute to are spread across the city. The transport options are many — metro, PMPML buses, autos, cabs, personal vehicles — but whether any of these is actually "good" for students is a different question. Cost, time, convenience, and how satisfied students feel about their daily commute are all things we wanted to understand.

This project is our attempt to answer those questions using data from 724 real survey responses from college students from the age group of 17-24 specifically. We did not want to just plot some graphs and move on — we wanted to actually understand what the data was telling us about how students in Pune travel, and whether there are things that could realistically be improved.

2. Objectives

The main things we were trying to figure out through this project were:

- Understand which modes of transport students most commonly use and why.
- Find out how much time students typically spend commuting one-way and if it varies a lot by transport mode.
- Look at the cost side — how much does a one-way commute cost and which modes are cheapest or most expensive.
- Identify the most common bottlenecks and delay points that affect student commutes.
- See if there is any real relationship between travel time, cost, and how satisfied students feel about their commute.
- Come up with some practical, realistic suggestions based on what the data shows.

3. Dataset Description

The data for this project was collected through a Google Form survey that we shared within the student community. We got 724 responses in total, which is a pretty decent sample size for a college-level project. Each response represents one student's daily commute profile.

The dataset has 10 columns covering different aspects of the commute. The key ones are the starting point of travel (areas like Kothrud, Baner, Wakad, Pashan, Shivajinagar, SIT Lavale Campus, etc.), the destination (Kharadi, Pune Station, Viman Nagar, Swargate, Hinjewadi, etc.), the primary mode of transport used (Metro, Cab/Ola/Uber, Carpool, Auto Rickshaw, Personal Two-Wheeler, PMPML City Bus), a secondary mode if the commute had multiple legs, the time of day they usually travel, average travel time in minutes, one-way cost in INR, the major delay points or bottlenecks they face, and their overall satisfaction score on a scale of 1 to 5.

We chose these parameters because they cover the most important dimensions of a student commute — time, money, location, and experience. The data was collected in March 2026 and captures real, ground-level travel behaviour rather than any simulated or assumed data.

4. Data Cleaning and Preparation

Raw survey data is almost never perfectly clean, and ours was no different. The first thing we did was load the dataset and check its basic structure — dimensions, column types, and how many values were missing.

We found around 132 missing values scattered across a few columns, mostly in the secondary mode and delay points fields, which makes sense since not everyone has a secondary mode and not everyone reported a delay point. We handled these by either filling them with a placeholder like "None" or leaving them as-is depending on the analysis we were doing.

We also inspected the column data types to make sure numeric columns like travel time and cost were actually being read as numbers, and categorical columns were consistent in their values (for example, making sure "Metro" was not appearing in multiple different forms due to typos).

On top of that, we did a bit of feature engineering — we created a new column called Commute_Category that labels each commute as Short (30 min or less), Medium (31–75 min), or Long (more than 75 min). This helped us group and compare commutes more meaningfully in some of the later analyses.

5. Exploratory Data Analysis

This section goes through the important visualizations we created and what we understood from each of them. We made a lot of graphs but have selected the most meaningful ones here.

5.1 Distribution of Transport Modes, Travel Time, Cost, and Peak Hours

The first set of charts gives us a broad picture of how the 724 students are distributed across different transport modes, what the travel times look like, how costs are spread, and when most students are commuting.

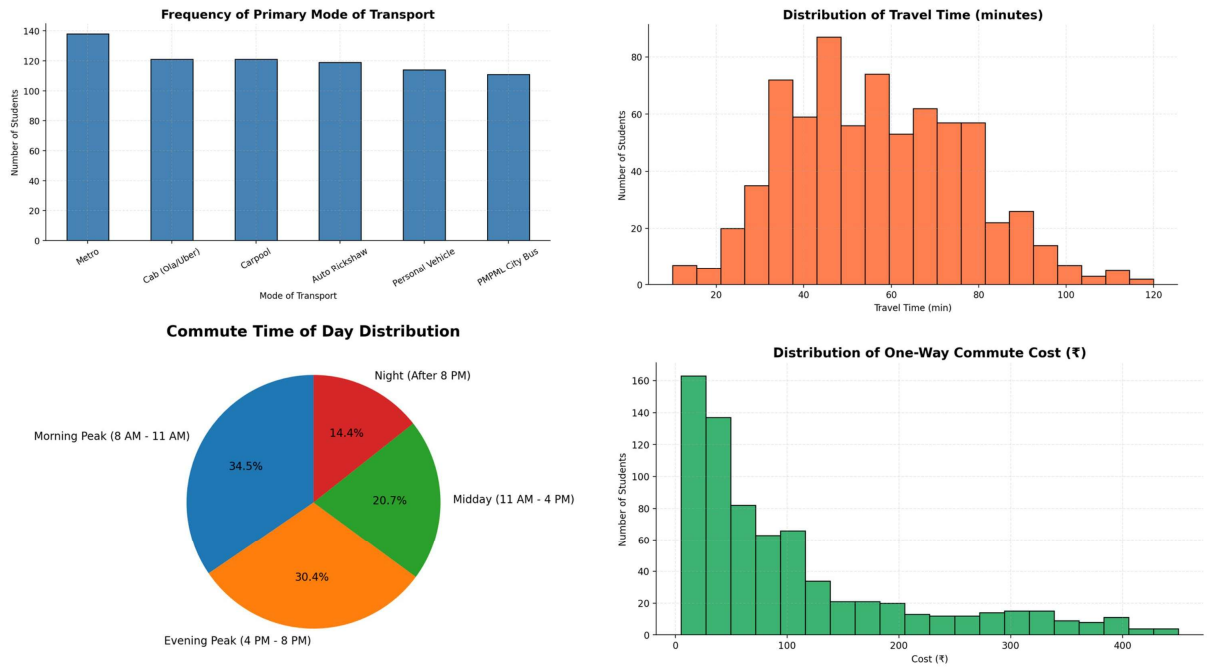

Figure 1 — Transport Mode Distribution, Travel Time Histogram, Peak Hours Distribution, and One-Way Cost Distribution

Looking at the transport mode bar chart, Metro is the most used mode with around 138 students, followed closely by Cab, Carpool, Auto Rickshaw, Personal Vehicle, and PMPML City Bus. What is interesting is that no single mode dominates overwhelmingly — usage is actually fairly balanced across all six, suggesting that students make varied choices based on their specific routes and situations.

The travel time histogram shows that most commutes fall between 40–60 minutes, which honestly is quite long. The distribution is roughly bell-shaped with a lean towards the higher side. Practically nobody is commuting in under 20 minutes, and a noticeable chunk goes beyond 80–90 minutes. That is a significant amount of time spent just getting to college and back.

The time-of-day pie chart shows morning peak (8–11 AM) is the dominant travel slot at 34.5%, followed by evening peak at 30.4% and midday at 20.7%. Night travel after 8 PM is a smaller segment at 14.4%, though still present — which could reflect late classes or part-time work situations.

The cost distribution histogram is right-skewed — most students pay under ₹100 for a one-way trip, but a few outliers go up to ₹400+. The bulk of students are clustered in the lower cost range, which reflects the large number of metro and bus users keeping costs down.

5.2 Average Travel Time and Cost by Transport Mode, Satisfaction Distribution, and Correlation

These four charts together tell us a lot about which transport modes are actually performing well and whether students are satisfied with their commutes.

Figure 2 — Average Travel Time by Mode, Average Cost by Mode, Satisfaction Distribution, and Correlation Heatmap

The average travel time chart is quite revealing. Metro is the fastest at around 39–40 minutes on average, while PMPML City Bus is the slowest at about 75 minutes. Carpool and Cab are somewhere in the middle, both around 60–65 minutes. So if speed is the priority, Metro is clearly the winner.

When you look at the cost chart, it flips the story significantly. Metro is the cheapest at roughly ₹23 on average, while Cab (Ola/Uber) is massively expensive at nearly ₹300 on average. Auto Rickshaw also comes in quite high at around ₹140. Carpool is a mid-range option at about ₹85. So basically, Cab users are paying almost 13 times more than Metro users for a commute that is not even that much faster.

The satisfaction distribution is interesting because the most common rating is 3 out of 5 — a neutral-to-okay response. Very few students rated their commute a 5, and a fair number gave it a 1 or 2. This tells us that while people are managing their commutes, very few of them are actually happy about it.

The correlation heatmap shows something surprising — travel time and satisfaction have a weak negative correlation (-0.073), and cost and satisfaction have almost no correlation at all (0.025). This means longer or more expensive commutes do not automatically mean lower satisfaction. Other factors, like how reliable the service is or how comfortable the ride is, seem to matter more.

5.3 Transport Mode Usage Patterns by Time of Day and Box Plot Analysis

These charts let us look at how commute behaviour shifts across different times of day, and how spread out the travel times and costs are for each mode.

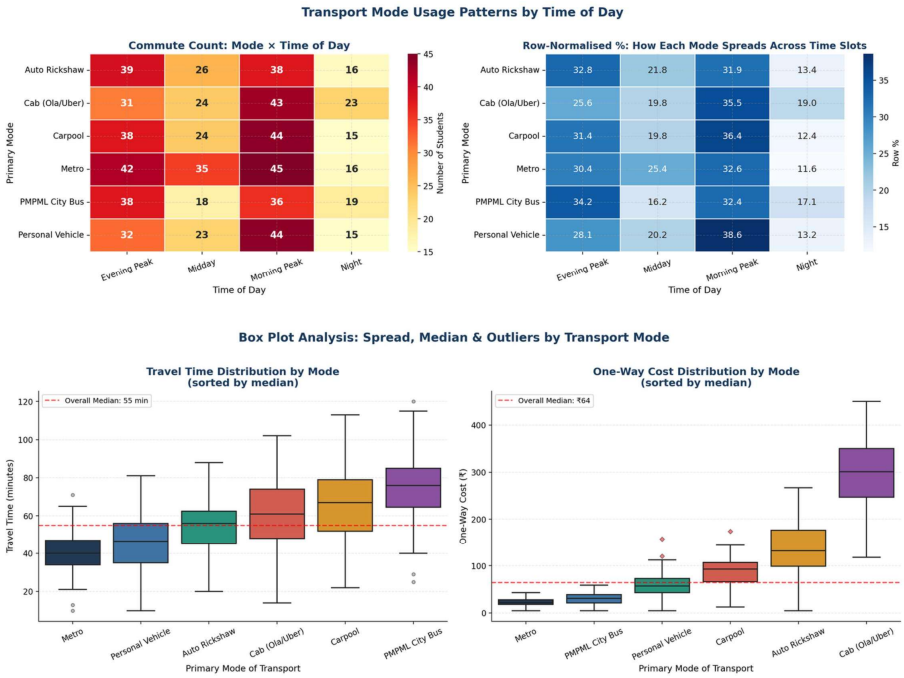

Figure 3 — Mode × Time of Day Heatmap and Box Plot Analysis for Travel Time and Cost by Mode

The crosstab heatmap (mode × time of day) shows that Metro usage peaks during morning commutes, which makes sense since it is the fastest option. Cab usage is relatively higher during evening peak hours — maybe because students are more willing to spend on convenience for the return journey, or public transport feels less reliable in the evening. PMPML Bus usage is actually spread fairly evenly across time slots, which is somewhat expected for a public service.

The box plots are great for understanding spread and outliers. For travel time, Metro has a tight distribution centered around 40 minutes — meaning it is not just fast on average, it is also consistent. PMPML City Bus has a much wider box and higher median, plus a few low outliers. For cost, Metro and PMPML Bus both show very low distributions with little spread, while Cab has a huge spread going up to ₹450+ in some cases. The Auto Rickshaw costs also show high variability, likely due to route-based pricing. This inconsistency in cab costs is worth noting — it means students relying on cabs cannot easily predict what they will spend.

5.4 Bubble Chart — Transport Mode Efficiency Map

This was probably our favourite chart to make because it puts travel time, cost, and satisfaction all in one visual, making comparisons really easy.

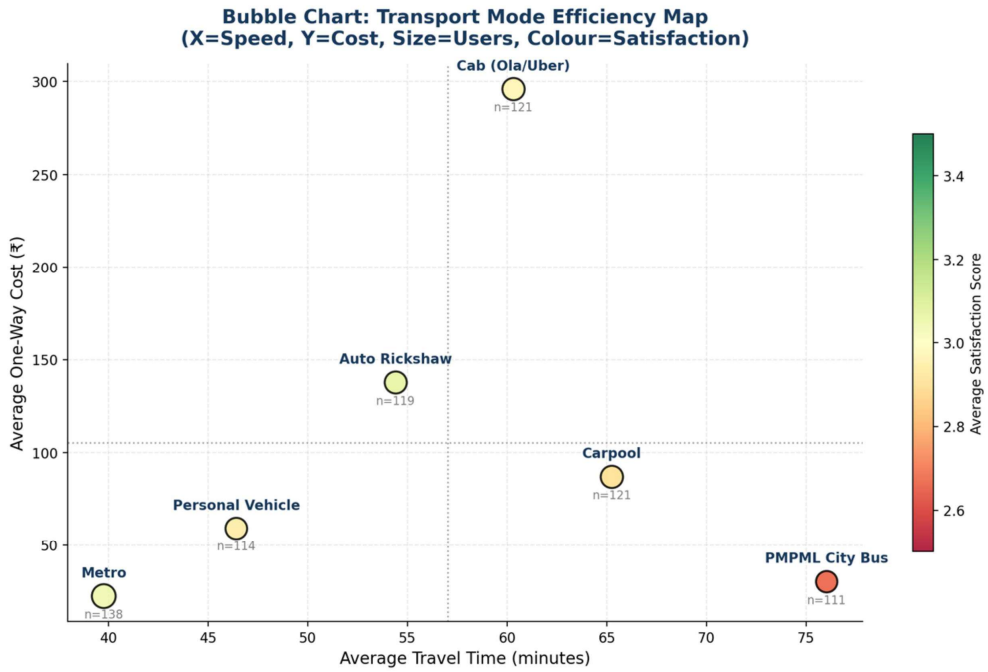

Figure 4 — Bubble Chart: Transport Mode Efficiency Map (X = Speed, Y = Cost, Size = Number of Users, Colour = Satisfaction)

The bubble chart maps each transport mode along two axes — average travel time on the X-axis and average one-way cost on the Y-axis. The size of the bubble shows how many students use that mode, and the colour reflects average satisfaction.

Metro sits in the bottom-left corner, which is the ideal zone — low time and low cost. It also has the largest user count (n=138). Cab is in the top-right, meaning expensive and slow compared to Metro. PMPML City Bus has the lowest cost after Metro, but it is the slowest of all modes and interestingly has the lowest satisfaction score (shown by the red colour). Personal Vehicle is a reasonable mid-zone option — not too expensive, decent speed. This chart really drives home why Metro is considered the best value option for student commuters.

5.5 Bottlenecks and Satisfaction Analysis

This slide focuses on what is actually causing delays and how satisfaction relates to travel time.

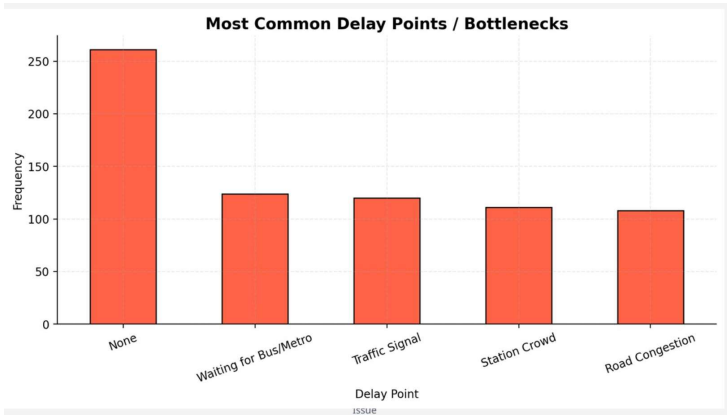

Figure 5 — Most Common Delay Points and Bottleneck Analysis

The bottleneck bar chart shows that "None" is the most reported value, meaning a good chunk of students do not face a specific identifiable delay. But among those who do face delays, "Waiting for Bus/Metro" is the single most common issue with 133 cases. This is a really important finding — students are not just stuck in traffic, they are waiting for public transport that is not frequent enough. Traffic signal delays and station crowding also show up frequently.

As we noted from the correlation heatmap, satisfaction is not strongly tied to travel time alone. The analysis from this slide reinforces that — Metro users rate their commute higher despite shared spaces and not drastically shorter times compared to some modes. What seems to matter more is predictability and service frequency. A student on a bus might take 70 minutes but if the bus is reliable, they can plan around it. If it is unpredictable, that is when frustration kicks in.

6. Key Observations

After going through all the analysis, here are the most important things we noticed:

- Metro is clearly the most efficient transport mode — fastest (~39 min average), cheapest (~₹23 average), most used (138 students), and highest satisfaction.
- Cabs are the most expensive option by a wide margin (~₹300 average one-way). Despite this, they do not offer dramatically faster travel times compared to other modes, which makes them a poor value-for-money choice.
- PMPML City Bus has the longest average travel time (~75 min) and the lowest satisfaction score among all modes. Despite being cheap, the reliability and time issues hurt user experience.
- The biggest commute pain point is "Waiting for Bus/Metro" — which points directly to a public transport frequency problem, not just a traffic or infrastructure problem.
- Travel time and cost are weakly correlated with satisfaction. The comfort and predictability of a commute matter more to students than raw time or money spent.
- Morning peak (8–11 AM) accounts for the highest volume of commutes at 34.5%, meaning the system is most stressed precisely when students need it most.
- Most commutes fall in the 40–60 minute range — a medium-length commute by any standard. Very few students have short commutes, which reflects the geographic spread of student residences across Pune.
- Students from Kothrud and Baner areas tend to rely more on two-wheelers and autos, while those near metro corridors heavily favour public transit.

7. Conclusion

Overall, this project gave us a much clearer picture of what daily student commuting looks like in Pune. Most students are spending close to an hour travelling one-way, which adds up to two hours a day just in transit. That is not a small amount when you are also expected to attend lectures, study, do assignments, and have some kind of personal life.

Metro has emerged as the undisputed winner for value and efficiency — it is fast, affordable, and students are generally more satisfied using it. The issue is that metro connectivity does not yet cover all student-heavy areas in Pune, which forces many to rely on slower or more expensive alternatives.

PMPML buses, while cheap, are let down by their frequency and travel time. Cabs are a convenience tax that most students probably cannot sustain daily. Two-wheelers and carpools occupy a reasonable middle ground for specific routes and distances.

What stood out the most was that satisfaction is not just about time or cost — it is about the overall reliability and experience of the commute. A student who knows exactly when their metro will arrive and can plan their day around it is going to be more satisfied than one who waits 30 minutes for a bus that is already packed when it shows up.

8. Possible Solutions and Recommendations

Based on what we found, here are some realistic suggestions — not just generic policy statements, but things that could actually help:

- Increase public transport frequency during peak hours (8–11 AM and 4–8 PM) — the biggest pain point is waiting, and this directly addresses it.
- Expand metro route coverage to areas like Kothrud, Baner, and Wakad that currently have lower metro access. These are high-density student zones with long average commute times.
- Improve last-mile connectivity from metro stations — a lot of students cannot use metro because there is no reliable way to get from the station to their final destination. Even e-rickshaws or dedicated shuttle services could help.
- Introduce student monthly passes for PMPML buses with real-time tracking apps — making the bus experience more predictable would significantly improve satisfaction without any infrastructure change.
- Colleges could facilitate official carpool groups for students from similar areas — it reduces cost, vehicles, and could be organized through a simple platform or WhatsApp group.
- Staggered class timings could help distribute the morning peak load — not every department needs to start at the same time.

9. Limitations

We want to be honest about what this project cannot claim:

- The data is self-reported, so travel times and costs are based on student estimates, not actual GPS or ticketing data. Some numbers might be off.
- 724 responses is decent for a college project but it is still a limited sample. It may not fully represent every student's experience across all of Pune.
- We only captured a snapshot in time — commute patterns may change seasonally, during exams, or with new infrastructure developments.
- The survey responses are skewed towards students who are aware of the project, which might introduce some selection bias.
- We did not account for variables like weather conditions, holidays, or special events that can significantly impact commute experience on any given day.
- Cost data does not account for shared-cost scenarios — for example, a car owner's actual daily cost is different from just fuel, and we may have had some inconsistencies in how students interpreted "one-way cost."

10. Key Findings Summary

The following table summarises the most significant findings from the EDA:

Finding	Detail
Total Survey Responses	724 students
Most Used Primary Mode	Metro (131 students, 18.09%)
Least Used Primary Mode	Personal Two-Wheeler (98 students, 13.54%)
Fastest Mode on Average	Metro (~38.68 min avg)
Slowest Mode on Average	PMPML City Bus (~81.47 min avg)
Cheapest Mode on Average	Metro (~Rs. 23 avg)
Most Expensive Mode on Average	Cab/Ola/Uber (~Rs. 298 avg)
Average Travel Time	57.47 minutes
Average One-Way Cost	Rs. 127
Most Common Starting Point	Kothrud (129 students)
Most Common Destination	Swargate & Viman Nagar (131 each)
Biggest Bottleneck	Waiting for Bus/Metro (133 cases)
Peak Commute Time Slot	Midday 11AM-4PM (205 commutes, 28.31%)
Most Common Satisfaction Score	Score 1 – Very Low (166 students)
Short Commutes (<=30 min)	109 students (15.06%)
Long Commutes (>60 min)	298 students (41.16%)
Missing Values Handled	132 nulls in Delay_Points — filled with 'None'